AI POWERED CONVERSATIONAL BI DASHBOARD

Turning Questions into Data Insights and Simplifying Business Intelligence with AI

Presented by Team Phoneix

INTRODUCTION

- Data analysis helps understand trends and patterns
- Businesses rely on visual insights
- Traditional tools can be complex
- Our system provides an easy analytics dashboard

TEAM MEMBERS

A GROUP OF STUDENTS FROM THE MCA 1ST YEAR PROGRAM UNDER MAKAUT (MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY).

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PROBLEM STATEMENT

- Users cannot write SQL queries
- Complex BI tools
- Time-consuming data analysis
- Lack of user-friendly systems

OBJECTIVE OF THE PROJECT

- Convert English queries to SQL
- Automate data analysis
- Generate visual insights
- Make BI tools user-friendly

TECHNOLOGY STACK

Frontend & Framework

STREAMLIT

Serves as BOTH frontend UI and backend server — no HTML/CSS/JS needed. Handles all user interactions.

PLOTLY EXPRESS

Renders interactive, zoomable charts directly in the browser. Supports 10+ chart types.

Backend & AI

PANDAS

Loads, filters, cleans, and transforms uploaded CSV datasets for analysis.

GEMINI AI ENGINE

LLM that parses natural language queries and recommends the ideal chart type + columns.

SQLITE

Lightweight database for user auth and for store data from given dataset.

KEY FEATURES

EVERYTHING YOU NEED FOR DATA INTELLIGENCE

CSV UPLOAD & AUTO-LOAD

NATURAL LANGUAGE INPUT

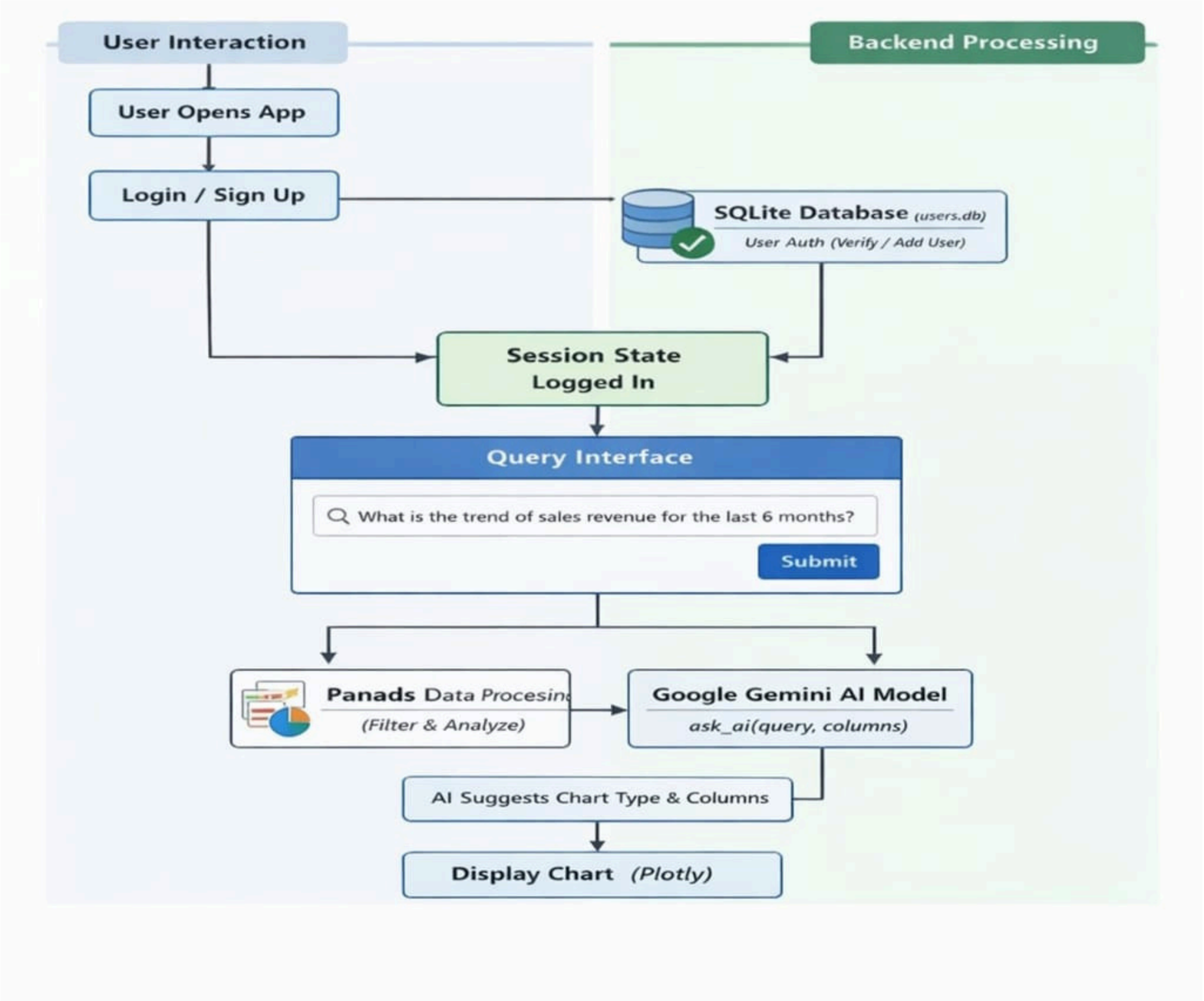
AI CHART RECOMMENDATION

INTERACTIVE VISUALIZATIONS

GENERATE QUERY& FILTERS DATASET

AUTH & SESSION MANAGEMENT

PROJECT WORKFLOW



ADVANTAGES OF THE PROJECT

- Easy to use
- Interactive visualization
- No need for complex web development
- Fast data analysis
- AI-based insights

WHAT COMES NEXT

- More Visualizations
- Geographic maps
- Support for multiple datasets
- Voice-based query input
- Multi-user collaboration
- Integration with real-time business data sources

CONCLUSION

- Built an interactive data dashboard
- Enables simple dataset analysis
- Provides visual insights through charts
- AI feature enhances data understanding

THANK YOU

We appreciate your attention to our presentation.